

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I E.Ragu / 1925710321 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ragu E

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug?
Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

Answer – Water Jug Problem

Initial State:

4-gallon jug = 0 gallons

3-gallon jug = 0 gallons

Step	Action	4-Gallon Jug	3-Gallon Jug
1	Fill the 3-gallon jug completely	0	3
2	Pour the 3 gallons into the 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full (only 1 gallon is needed)	4	2
5	Empty the 4-gallon jug onto the ground	0	2
6	Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug	2	0

Goal Achieved:

The 4-gallon jug now contains exactly 2 gallons of water.

Conclusion:

By filling, transferring, and emptying the jugs in the sequence above, we successfully obtain exactly 2 gallons in the 4-gallon jug.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?

Answer:

i. What are the percepts for this agent?

Percepts are the inputs received by the Mars Rover from its environment through sensors.

- Images from cameras
- Surface and terrain data
- Rock and soil composition data
- Temperature readings
- Atmospheric pressure measurements
- Wind speed and direction
- Position and navigation information
- Obstacle detection data

ii. Characterize the operating environment

The Mars Rover operates in:

- Partially Observable: The rover cannot observe the entire environment at once.
- Deterministic: Most actions have predictable outcomes, though terrain conditions may introduce uncertainty.
- Sequential: Current actions affect future states.
- Dynamic: Environmental conditions such as dust storms may change.
- Continuous: Movement, time, and sensor readings are continuous.
- Single-Agent: The rover performs tasks independently.

iii. What are the actions the agent can take?

The Mars Rover can:

- Move forward, backward, left, or right
- Navigate around obstacles
- Capture images and videos
- Drill rocks and collect samples
- Analyze soil and rock composition
- Transmit data to Earth
- Monitor weather conditions
- Recharge and manage power resources

iv. How can one evaluate the performance of the agent?

Performance measures include:

- Accuracy of scientific observations
- Number of useful samples collected
- Distance covered successfully
- Successful obstacle avoidance

v. What sort of agent architecture is most suitable? Why?

Most Suitable Architecture: Goal-Based Agent with Model-Based Features.

Reason:

- The rover has specific mission goals such as exploration and sample collection.
- It must maintain an internal model of the environment.
- It uses sensor information to make decisions.
- It plans actions to achieve goals while avoiding obstacles and conserving energy.

Therefore, a Goal-Based Model Agent is the most suitable architecture for a Mars Rover because it can plan intelligently and adapt to changing environmental conditions.

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

Answer:

Place 8 queens on an 8×8 chessboard so that no queen can attack another queen.

A queen can attack:

- Horizontally (same row)
- Vertically (same column)
- Diagonally

The objective is to place all 8 queens without any conflicts.

Problem Components

1. Initial State

An empty chessboard with no queens placed.

State:

```
[ ]  
[ ]  
[ ]  
[ ]  
[ ]  
[ ]  
[ ]
```

2. State Space

A state represents a partial arrangement of queens on the board.

Example:

```
Q.....  
...Q....  
.Q.....  
.....Q..  
..Q.....  
.....Q.  
....Q....  
.....
```

Each state is generated by placing one queen in a valid position.

3. Actions (Operators)

- Place a queen in a safe square of the next column.
- Ensure that the new queen is not attacked by previously placed queens.

4. Goal Test

The goal is achieved when:

- 8 queens are placed on the board.
- No two queens share the same row.
- No two queens share the same column.
- No two queens share the same diagonal.

5. Path Cost

Each queen placement costs 1.

Total path cost = 8 (placing 8 queens).

Search Strategy

Backtracking Search

1. Place a queen in the first column.
2. Move to the next column and place another queen in a safe position.
3. If no safe position exists, backtrack to the previous column.
4. Continue until all 8 queens are placed successfully.

One Valid Solution

Column : 1 2 3 4 5 6 7 8

Row : 1 5 8 6 3 7 2 4

Board Representation:

```
Q.....
.....Q.
....Q...
.....Q
.Q.....
...Q....
.....Q..
..Q.....
```

No two queens attack each other.

Conclusion

The 8-Queens problem is a classical AI search problem. It is solved efficiently using Backtracking Search, where queens are placed column by column and invalid states are eliminated until a valid arrangement of all 8 queens is obtained.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

Answer:

Type of Problem-Solving Agent

Goal-Based Agent

Reason:

A Goal-Based Agent is suitable because the customer has a specific goal:

- Reach the destination safely and comfortably.
- Select a cab type based on preferences such as cost, comfort, and availability.
- The agent evaluates available options and chooses the best action to achieve the goal.

Problem Formulation

Initial State

- Customer opens the OLA application.
- Current location is identified.

Actions

- Enter destination.
- View available cab types.
- Compare fare estimates.
- Select a cab type.
- Confirm booking.
- Track cab and complete journey.

Goal State

- Customer successfully reaches the destination.

Path Cost

- Fare amount.
- Travel time.
- Waiting time.

Pseudocode

BEGIN

Open OLA Application

INPUT current_location

INPUT destination

Display available cab types

(Mini, Micro, Sedan, Prime, Shared)

FOR each cab type

Display fare estimate

Display arrival time

END FOR

User selects preferred cab

IF cab is available THEN

Confirm booking

Assign driver

Track ride

```
Travel to destination
Complete payment
Display "Destination Reached Successfully"
ELSE
Display "Cab Not Available"
ENDIF
```

END

PEAS Description

Performance Measure

- Minimum travel cost
- Reduced waiting time
- Customer satisfaction
- Safe arrival at destination

Environment

- Roads, traffic, GPS network, drivers, passengers

Actuators

- Booking request
- Route selection
- Notifications
- Payment processing

Sensors

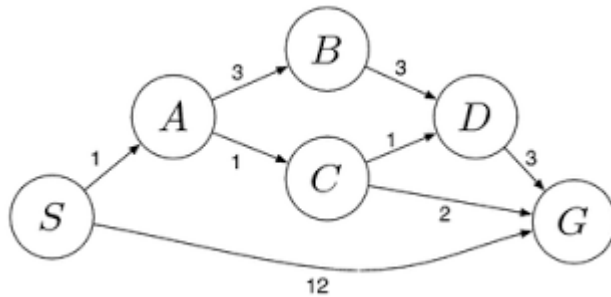
- GPS
- User location
- Traffic data
- Driver availability

Conclusion

The OLA Cab Booking system is best modeled as a **Goal-Based Agent** because it selects actions that help the customer achieve the goal of reaching the destination efficiently and comfortably.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse **G** using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Answer:

Problem Statement

A logistics company wants to find the least-cost path from the starting warehouse **S** to the destination warehouse **G**. Each route has an associated travel cost.

Uniform Cost Search (UCS) is used to expand the node with the lowest cumulative cost first.

UCS Algorithm

Steps

1. Start from node **S** with cost 0.
2. Insert **S** into the priority queue.
3. Remove the node with the lowest path cost.
4. If the node is the goal node **G**, stop.
5. Otherwise, expand the node and add its neighbors with cumulative costs.
6. Repeat until **G** is reached.

Pseudocode

UCS(Graph, Start, Goal)

Create PriorityQueue PQ

Insert (Start, 0) into PQ

While PQ is not empty

Remove node with minimum cost

If node = Goal
Return path and cost

Expand node

For each neighbor
 $\text{NewCost} = \text{CurrentCost} + \text{EdgeCost}$

If neighbor not visited
Add neighbor to PQ with NewCost

End While

Advantages of UCS

- Guarantees the least-cost solution.
- Suitable for transportation and logistics networks.
- Considers route cost rather than number of steps.

Conclusion

Uniform Cost Search explores nodes in increasing order of cumulative path cost and always finds the optimal (minimum-cost) path from warehouse S to warehouse G.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process

Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem